

In this thought experiment, Schrödinger imagined a cat inside a box with a device that has a 50% chance of releasing poison. According to quantum theory, until the box is opened and the cat is observed, the cat is both alive and dead at the same time. This strange idea helps explain superposition in quantum computing—where a qubit can be in a state of 0, 1, or both at once, which is different from a regular computer bit that can only be 0 or 1.

In the 1980s, scientists like Richard Feynman and David Deutsch began discussing how quantum mechanics could be used to build computers capable of solving problems much faster than traditional computers. They suggested that quantum computers could handle complex calculations and vast amounts of data at unimaginable speeds, potentially leading to groundbreaking advancements.

A significant breakthrough in the field of quantum computing came in 1994 when mathematician Peter Shor introduced Shor's Algorithm. This algorithm demonstrated that quantum computers could factor large numbers exponentially faster than classical computers. Factoring large numbers means breaking them down into smaller prime numbers. For example, the number 15 can be factored into 3 and 5, which are prime numbers that multiply to give 15. However, as the numbers grow much larger—such as those used in modern encryption—it becomes incredibly difficult and time-consuming for classical computers to factor them. This process is essential for many encryption techniques that protect sensitive information online, such as credit card details and private communications.

Shor's discovery showed that quantum computers could perform this task at a much faster rate, potentially breaking encryption methods that rely on the difficulty of factoring large numbers. This posed a significant threat to data security, as current encryption systems might become vulnerable to

quantum attacks. As a result, his work sparked a global effort to advance quantum computing research and develop new encryption methods capable of resisting the power of quantum computers.

In 2019, Google claimed to have achieved quantum supremacy, where its quantum computer completed a task in 200 seconds that would take classical supercomputers 10,000 years. This milestone signalled the potential of quantum technology to revolutionise industries. However, IBM disagreed, arguing that the task could be completed in just 2.5 days using a traditional computer, and urged caution in accepting Google's claim. Google defended its achievement, comparing it to the Wright brothers' first flight, which showed the possibility of flight, even if it had no immediate practical use. Despite the debate, quantum computing is still evolving.

The quantum future – opportunities and challenges

Quantum computing has the potential to revolutionise many industries, from finance to healthcare, and beyond. One of its most exciting applications is in cryptography and cybersecurity. Today, we rely on encryption methods to protect our personal and financial data online. However, as quantum computers become more powerful, they could easily break these encryption systems. This is why quantum-safe encryption—methods that are secure against quantum attacks—is becoming crucial. In India, companies like QNu Labs are already developing quantum-safe encryption technologies, while global giants like IBM and Microsoft are also working on similar solutions to secure data.

In healthcare, quantum computers have the potential to change the way we develop medicines. They can simulate molecules and chemical reactions at a level of detail that

classical computers simply can't match. This ability could speed up the process of discovering new drugs. For example, biotech companies in the US are using quantum computers to simulate proteins, which may lead to breakthroughs in the treatment of diseases like cancer and Alzheimer's.

In finance, quantum computing could make a huge difference by helping banks and investment firms solve complex problems faster and more accurately. JPMorgan Chase, in collaboration with IBM, is exploring how quantum computing could improve financial models, optimise investment portfolios, and predict risks more efficiently. In India, banks and financial institutions are following quantum developments closely, preparing for a future where quantum computing could change the way they do business—by enhancing fraud detection, improving trading algorithms, and optimising asset management.

The logistics and manufacturing industries are also set to benefit from quantum computing. Companies involved in transportation can use quantum algorithms to find the most efficient routes for their deliveries, saving fuel and reducing costs. Volkswagen, for example, has experimented with quantum computing to optimise traffic flow in cities, making transportation smoother and faster. Similarly, logistics companies in India are exploring how quantum computing can improve delivery networks, making them more efficient and cost-effective.

Another exciting area is AI. Quantum computing could take machine learning to the next level. AI systems learn from data, and quantum computers can process large datasets much faster and more efficiently than classical computers. Google is already researching how quantum computers could improve AI models for tasks like natural language processing, image recognition, and more. This could lead to smarter, faster, and more accurate AI systems in many fields.